# A ONE-DIMENSIONAL POINT OF A NORMALIZATION OUTSIDE THE IMAGE OF THE SOURCE HAS DISCRETE VALUATION RING

THE STACKS PROJECT AUTHORS

*Editorial note.* All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file pinned-original-source.tex; no statement or proof step was rewritten, reordered or removed.

*Editorial note.* Following the multi-claim convention for this source, source_url is the source-level URL https://stacks.math.columbia.edu/ and the source label lemma-relative-normalization-normal-codim-1 is carried in claim_key and here rather than in a per-claim tag URL.

*Editorial note.* The preamble is the chapter's shared Stacks Project preamble, retained as sources/stacks-morphisms/retained-preamble.tex: a line-for-line excerpt of the shared Stacks preamble staged with this snapshot. No macro was invented or redefined, and this card adds no preamble text of its own.

*Editorial note.* Statement span 14292-14302 and proof span 14304-14348 come from the halfA pool card operations/pool-cards/stacks-morphisms-halfA.json and were re-read in the pinned chapter file before this card was written.

*Editorial note.* Re-checked by hand before carding: this candidate was caught by the lane's thin-proof prefilter, and reading the original shows the hit comes only from the closing phrase 'Some details omitted.' while the proof is a full reduction argument (finiteness of the normalization, prime avoidance, vanishing-locus analysis, integral-closure computation). Per the fleet rule of 2026-10-14 the prefilter hit on a proof of more than 800 bytes was re-read by hand, and the reason is recorded here. The author's closing phrase is kept verbatim in the counted span.

*Editorial note.* 7 cross-references occur inside the counted spans; they are listed in external_references as navigation pointers only.

*Allowed standard prerequisites (editorial).*

- For a Nagata base and a finite type morphism with reduced source the normalization in the source is finite, cited from the same chapter. Cited at proof lines 14299-14301. Makes the normalization affine and finite over an affine base, so that the situation becomes a finite algebra over a Noetherian ring.
- The idempotence of the normalization, established earlier in the same chapter (and by a previous card of this lane), which lets the base be replaced by the normalization itself. Cited at proof lines 14301-14303. Reduces to the case where base and normalization coincide.
- Prime avoidance together with the finiteness of the set of minimal primes of a Noetherian ring, and the description of the closure of the image of a closed subset as the specializations of its points. Cited at proof lines 14306-14309 and 14318-14320. Produce the element avoiding the minimal primes and show that the closure of the image of its vanishing locus avoids the point.

*Date*: stacks-project:morphisms.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

- The compatibility of integral closure with localization, and the characterization of a reduced ring as normal via its local behaviour, cited from the algebra chapter. Cited at proof lines 14333-14335 and 14342-14345. Turn the equality between the ring and the integral closure in a localization into normality of the local ring at the point.

*Source reference.* Original lemma 00FR: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 581–581. Same-chapter reference used in the retained author text at source line(s) 13632, 14321, 15637. It is supplied only as a navigation pointer so that the retained text compiles; the counted item delivers the author's own statement and the author's complete proof, and no part of the referenced item is copied in or substituted for the counted argument.

*Source reference.* Original lemma 030C: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 3943–3943. Same-chapter reference used in the retained author text at source line(s) 14346, 14454. It is supplied only as a navigation pointer so that the retained text compiles; the counted item delivers the author's own statement and the author's complete proof, and no part of the referenced item is copied in or substituted for the counted argument.

*Source reference.* Original lemma 0307: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 3938–3938. Same-chapter reference used in the retained author text at source line(s) 13874, 14339, 14493. It is supplied only as a navigation pointer so that the retained text compiles; the counted item delivers the author's own statement and the author's complete proof, and no part of the referenced item is copied in or substituted for the counted argument.

*Source reference.* Original lemma 00DS: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 508–508. Same-chapter reference used in the retained author text at source line(s) 14320. It is supplied only as a navigation pointer so that the retained text compiles; the counted item delivers the author's own statement and the author's complete proof, and no part of the referenced item is copied in or substituted for the counted argument.

*Source reference.* Original lemma 03GR: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 4538–4538. Same-chapter reference used in the retained author text at source line(s) 14307. It is supplied only as a navigation pointer so that the retained text compiles; the counted item delivers the author's own statement and the author's complete proof, and no part of the referenced item is copied in or substituted for the counted argument.

*Source reference.* Original lemma 0BXA: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 14973–14973. Same-chapter reference used in the retained author text at source line(s) 14310. It is supplied only as a navigation pointer so that the retained text compiles; the counted item delivers the author's own statement and the author's complete proof, and no part of the referenced item is copied in or substituted for the counted argument.

*Source reference.* Original lemma 02JQ: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 3345–3345. Same-chapter reference used in the retained author text at source line(s) 1367, 4438, 10265, 14326. It is supplied only as a navigation pointer so that the retained text compiles; the counted item delivers the author's own statement and the author's complete proof, and no part of the referenced item is copied in or substituted for the counted argument.

**Lemma 0.1.** *Let $f : Y \to X$ be a finite type morphism of schemes with $Y$ reduced and $X$ Nagata. Let $X'$ be the normalization of $X$ in $Y$. Let $x' \in X'$ be a point such that*

(1) $\dim(\mathcal{O}_{X',x'}) = 1$, *and*
(2) *the fibre of $Y \to X'$ over $x'$ is empty.*

*Then $\mathcal{O}_{X',x'}$ is a discrete valuation ring.*

*Proof.* We can replace $X$ by an affine neighbourhood of the image of $x'$. Hence we may assume $X = \mathrm{Spec}(A)$ with $A$ Nagata. By Lemma 03GR the morphism $X' \to X$ is finite. Hence we can write $X' = \mathrm{Spec}(A')$ for a finite $A$-algebra $A'$. By Lemma 0BXA after replacing $X$ by $X'$ we reduce to the case described in the next paragraph.

The case $X = X' = \mathrm{Spec}(A)$ with $A$ Noetherian. Let $\mathfrak{p} \subset A$ be the prime ideal corresponding to our point $x'$. Choose $g \in \mathfrak{p}$ not contained in any minimal prime of $A$ (use prime avoidance and the fact that $A$ has finitely many minimal primes, see Algebra, Lemmas 00DS and 00FR). Set $Z = f^{-1}V(g) \subset Y$; it is a closed subscheme of $Y$. Then $f(Z)$ does not contain any generic point by choice of $g$ and does not contain $x'$ because $x'$ is not in the image of $f$. The closure of $f(Z)$ is the set of specializations of points of $f(Z)$ by Lemma 02JQ. Thus the closure of $f(Z)$ does not contain $x'$ because the condition $\dim(\mathcal{O}_{X',x'}) = 1$ implies only the generic points of $X = X'$ specialize to $x'$. In other words, after replacing $X$ by an affine open neighbourhood of $x'$ we may assume that $f^{-1}V(g) = \emptyset$. Thus $g$ maps to an invertible global function on $Y$ and we obtain a factorization

$$A \to A_g \to \Gamma(Y, \mathcal{O}_Y)$$

Since $X = X'$ this implies that $A$ is equal to the integral closure of $A$ in $A_g$. By Algebra, Lemma 0307 we conclude that $A_\mathfrak{p}$ is the integral closure of $A_\mathfrak{p}$ in $A_\mathfrak{p}[1/g]$. By our choice of $g$, since $\dim(A_\mathfrak{p}) = 1$ and since $A$ is reduced we see that $A_\mathfrak{p}[1/g]$ is a finite product of fields (the product of the residue fields of the minimal primes contained in $\mathfrak{p}$). Hence $A_\mathfrak{p}$ is normal (Algebra, Lemma 030C) and the proof is complete. Some details omitted. □